

Application 1: p22 ch3

. moveud E (Max):

$$V = -\infty, \text{ Fibs} = L/M/N$$

$$V = \text{Max}(-\infty, -3) = -3$$

$$V = \text{Max}(-3, 15) = 15$$

$$V = \text{Max}(15, -5) = 15$$

$$\text{MinMax}(E) = 15$$

. moveud F (Max):

$$V = -\infty, \text{ Fibs} = O/P$$

$$V = \text{Max}(-\infty, -10) = -10$$

$$V = \text{Max}(-10, 3) = 3$$

$$\text{MinMax}(F) = 3$$

. moveud G (Max):

$$V = -\infty, \text{ Fibs} = Q/R$$

$$V = \text{Max}(-\infty, 4) = 4$$

$$V = \text{Max}(4, 6) = 6$$

$$\text{MinMax}(G) = 6$$

. moveud H (Max):

$$V = -\infty, \text{ Fibs} = S/T$$

$$V = \text{Max}(-\infty, -7) = -7$$

$$V = \text{Max}(-7, 12) = 12$$

$$\text{MinMax}(H) = 12$$

. moveud I (Max)

$$V = -\infty, \text{ Fibs} = U/V$$

$$V = \text{Max}(-\infty, 4) = 4$$

$$V = \text{Max}(4, 8) = 8$$

$$\text{MinMax}(I) = 8$$

. moveud J (Max):

$$V = -\infty, \text{ Fibs} = W/X/Y$$

$$V = \text{Max}(-\infty, 20) = 20$$

$$V = \text{Max}(20, 12) = 20$$

$$V = \text{Max}(20, 30) = 30$$

$$\text{MinMax}(J) = 30$$

. moveud K (Max):

$$V = -\infty, \text{ Fibs} = Z/A$$

$$V = \text{Max}(-\infty, 20) = 20$$

$$V = \text{Max}(20, 10) = 20$$

$$\text{MinMax}(K) = 20$$

• module B (Min)

$$V = +\infty, \text{ Fib} = E/F$$

$$V = \text{Min} (+\infty, 15) = 15$$

$$V = \text{Min} (15, 3) = 3$$

$$\text{Min Max} (B) = 3$$

• module C (Min)

$$V = +\infty, \text{ Fib} = G/H$$

$$V = \text{Min} (+\infty, 6) = 6$$

$$V = \text{Min} (6, 12) = 6$$

$$\text{Min Max} (C) = 6$$

• module D (Min)

$$V = +\infty, \text{ Fib} = (I/J)K$$

$$V = \text{Min} (+\infty, 8) = 8$$

$$V = \text{Min} (8, 30) = 8$$

$$V = \text{Min} (8, 20) = 8$$

$$\text{Min Max} (D) = 8$$

Alpha-Beta vs MinMax p32 ch3

Séance 2

= 6 points (1)

- Niveau 1 = a (Max)
- Niveau 2 = (b, c, d) (Min)
- Niveau 3 = les feuilles (e, f, ..., m)
- α = Meilleure valeur que Max peut garantir ($-\infty$)
- β = " " que Min peut garantir ($+\infty$)
- Elégance = on arrête l'exploration si $\alpha \geq \beta$

• Etape 1 = Branche (b) (Min)

b (Min), $\alpha = -\infty$, $\beta = +\infty$

les fils, e = 7, f = 3, g = 5

• e = 7 $\rightarrow \beta = \min(+\infty, 7) = 7$

$\alpha = -\infty < \beta(7)$, on continue.

• f = 3 $\rightarrow \beta = \min(7, 3) = 3$

$\alpha = -\infty < 3$, on continue.

• g = 5 $\rightarrow \beta = \min(3, 5) = 3$

$\alpha = -\infty < 3$, on continue.

• Etape 2 =

$\alpha = \max(-\infty, 3) = 3$

$\alpha = 3$

• Etape 3:

$$C(\text{Min}) : \alpha = 5, \beta = +\infty$$

$$h = 4 \rightarrow \beta = \min(+\infty, 4) = 4$$

$$\alpha = 5 > \beta = 4$$

- condition d'élégage:

• Min choisit une valeur ≤ 4

• $\text{Max}(a) = a$ déjà meilleur option ($\alpha = 5$ via b)

$\Rightarrow$ on n'explore pas $i = 10, j = 2$

$$a = \text{Max}(5, 4) = 5$$

• Etape 4: Branche (d) Min.

$$\alpha = 5, \beta = +\infty$$

les fils: $k = 7, l = 3, m = 9, n = 1$

$$k = 7 \rightarrow \beta = \min(+\infty, 7) = 7$$

$$\alpha = 5 < \beta = 7, \text{ on continue.}$$

$$l = 3 \Rightarrow \beta = \min(7, 3) = 3$$

$$\alpha = 5 \geq \beta = 3$$

- condition d'élégage:

• Min choisit une valeur ≤ 3

• $\text{Max}(a)$ a déjà meilleur option ($\alpha = 5$)

Cette branche est fine pour Max que 5

donc on élégage

$\Rightarrow$ on n'explore pas $m = 9, n = 1$

d retourne 3

$$a = \text{Max}(3, 5) = 5, \alpha = 5$$

Exemple p 35

$$\alpha = -\infty$$

$$\beta = +\infty$$

$$V = -\infty$$

* Etape 1: T2 (Min)

$$\alpha = -\infty, \beta = +\infty$$

Les fils (D, E)

$$D \Rightarrow \beta = \min(+\infty, 3) = 3$$

$$E \Rightarrow \beta = \min(3, 5) = 3$$

La valeur finale de T2 est 3

* De retour à T1 (Max)

$$\alpha = \max(-\infty, 3) = 3$$

* Etape 2: on va explorer T3

$$\alpha = 3, \beta = +\infty$$

. Noeud F: $\alpha = 3, \beta = +\infty$

. Noeud I: $\alpha = 3, \beta = +\infty$

. Noeud Q: 0

$$\beta = \min(+\infty, 0) = 0$$

$\beta < \alpha (0, 3) \Rightarrow$ Il faut élaguer R. S. T

* Retour à F: $\alpha = \max(3, 0) = 3$

* Noeud J (Min): $\alpha = 3, \beta = +\infty$

$$U \rightarrow 5, \beta = \min(+\infty, 5) = 5$$

$$V \rightarrow 6, \beta = \min(5, 6) = 5$$

* Retour à F:

$$\text{Max}(3, 5) = 5$$

* Retourne à T3:

$$\beta = \min(+\infty, 5) = 5$$

. Noeud G:

$$\alpha = 3, \beta = 5$$

. Noeud W: 7

$$\alpha = 3, \beta = +\infty$$

$$\beta = \min(+\infty, 7) = 7$$

. Noeud X: 8

$$\beta = \min(7, 8) = 7$$

Valeur K = 7

* Retourne à G

$$\alpha = \max(3, 7) = 7$$

* $\alpha = 7 \geq \beta = 5 \Rightarrow$ élaguer L, M, N

. Noeud H

$$\alpha = 3, \beta = 5$$

. Noeud Q:

$$\text{Noeud AB: } \beta = \min(+\infty, 4) = 4$$

$$\text{Noeud A7: } 5$$

$$\beta = \min(4, 5) = 4$$

La valeur de Q = 4

* Retourne à H,

$\alpha = \max(4, 2) \Rightarrow$ La valeur finale de H = 4

Retourne à T3 =

$$B = \min(5, 7, 4) = 4$$

* Retour à la racine T1

$$\text{Max}(3, 4) = 4$$

La valeur finale de la racine = 4

⇒ Le joueur Max peut garantir un score de 4 en jouant de manière ~~général~~ optimale, en supposant que Min joue aussi de manière optimale.

T2 (Min)

$$V = +\infty, \alpha = -\infty, \beta = +\infty$$

$$V = -\infty$$

$$\text{Max}(\text{AlphaBeta}(D, \alpha, \beta))$$

$$\text{Min}(+\infty, 3) = 3$$

$$\beta = \text{Min}(\beta, V) = 3$$

$$V = 3$$

$$\beta = 3$$

$$E = \text{AlphaBeta}(E, \alpha, \beta) = 5$$

$$\text{Min}(3, 5) = 3$$

$$\beta = 3$$

$$\text{T2}(\alpha = -\infty, 3)$$

$$V = 3$$

T1 (Max)

$$V = -\infty$$

$$\alpha = \text{Max}(3, -\infty) = 3$$

$$\alpha = 3$$

$$\beta = +\infty$$

$$V = 3$$

$$I: \alpha = 3, \beta = +\infty$$

$$V = -\infty$$

$$V = \text{Min}(V, \text{AlphaBeta}(Q)) = 0$$

$$V = 0 \leq 3 = \alpha$$

$$\beta = \text{Min}(+\infty, 0) = 0$$

$$I = \text{Max}(3, 0) = 3$$

$$V = 0$$

$$\beta = +\infty$$

$$F = 3, \alpha$$

$$\beta = +\infty$$

$$V = -\infty \text{Max}(-\infty, 0) = 0$$

$$\text{AlphaBeta}(U) = 5$$

$$V = +\infty$$

$$V = \text{Min}(+\infty, 5) = 5$$

$$\beta = \text{Min}(+\infty, 5) = 5$$

$$V = 5$$

$$\alpha = 3 \leq 5$$

$$\text{AlphaBeta}(V = 2)$$

$$V = \text{Min}(5, 6) = 5$$

$$V > \alpha$$

$$\beta = \text{Min}(5, 6) = 5$$

Retourne à F:

$$\alpha = 3$$

$$\beta = 5$$

$$V = 5$$

$$\alpha = \text{Max}(3, 5) = 5$$

T3 (lim)

$$\beta: \lim(+\infty, 5) = 5$$

$$V = 5$$

$$\begin{matrix} \alpha & \beta \\ \rightarrow & (3, 5) \end{matrix}$$

$$V = 5$$

$$G: \alpha = 3$$

$$\beta = 5$$

$$V = -\infty$$